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# Object detection algorithms for car tracking with euclidean distance tracking and YOLO

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## Abstract

Traffic congestion is an important problem that causes wasted gasoline, lost time, and stress. Traffic congestion is caused by the high number of motorized vehicles on the road and ineffective traffic light control. Dynamic traffic lights are a solution to reduce traffic congestion by changing the duration of the lights based on vehicle density. This research uses object detection algorithms for car tracking and finding the most effective algorithm. The result of this work can be utilized in real-time analysis of traffic conditions by detecting and tracking vehicles at crossroads. This work tests and compares Euclidean Distance Tracking and YOLO algorithms for tracking moving objects. The testing dataset is traffic videos from highway CCTV footage. The datasets were pre-processed and passed to the tracking algorithms. Lastly, the performance of both algorithms is compared. The experiment shows that the Euclidean algorithm performs worse than YOLO. Euclidean algorithm erroneously identifies certain parts of the vehicle, such as the windows, rear-view mirrors, and even the vehicle's shadows, as part of the vehicle itself. In contrast, YOLO can effectively track vehicles with minimal errors. This study also highlights limitations in accurately tracking and counting vehicles using the current system and video footage. For example, the tracking performance was affected in scenarios involving camera movement or multiple vehicle types.

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**Keywords:** dynamic traffic light, track vehicles, distance tracking, traffic congestion, euclidean distance, YOLO

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## 1. Introduction

Traffic congestion and monitoring are among the most critical issues worldwide [1, 2]. The large number of vehicles creates an imbalance between the demand and supply of transportation in traffic infrastructure, which leads to problems such as increased traffic accidents, travel delays, and environmental damage. In 1912, traffic lights were invented for the first time, which were used to accommodate vehicles effectively so that intersections became more

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orderly [3]. However, there were still many problems with traffic jams due to the vehicular revolution. Moreover, ineffective traffic light control at intersections can lead to traffic jams. Traffic jams on a large scale can hinder motorists from being on time and increasing stress. To manage intersections, crossroads are often monitored and controlled by a traffic light control system with predetermined timings [4]. However, using standard traffic lights with static timing proved ineffective and can cause traffic congestion [5]. This issue occurs when a traffic light remains green for one direction at an intersection despite the road being unoccupied for an extended period.

A recent solution to reduce traffic density is installing dynamic traffic lights [4, 6]. These systems help by adjusting the duration of each traffic light based on real-time traffic conditions. For instance, if one of the sides in the intersection has more density than the other side, the dynamic traffic light could help increase or decrease the duration of the traffic light. As a result, it reduces the waiting line for the traffic light and reduces the traffic congestion [7]. Dynamic traffic lights work and require *car tracking algorithms*, which help detect the cars' density on each side of the intersection and send data to the traffic light. Generally, Internet of Things (IoT) devices and object detection algorithms are utilized to detect cars and understand traffic conditions [8].

This study aims to implement car-tracking systems using object detection algorithms, i.e., Euclidean distance tracking and YOLO. Then, it evaluates and compares those algorithms to determine which is more effective for car-tracking systems. The car detection algorithm from this study can be utilized to implement dynamic traffic lights to reduce traffic congestion.

This study's methodology starts by collecting a dataset of highway CCTV footage. Each video was pre-processed before being used for object detection algorithms. Then, this paper implements the Euclidean distance tracking algorithm and YOLO algorithm to track moving vehicles (i.e., cars, motorcycles, and trucks) in the video dataset. The experiment shows that the Euclidean algorithm falsely detects a vehicle's parts as a complete car. On the other hand, YOLO can track vehicles accurately and detect cars with minor errors.

The structure of the paper is as follows. Chapter 1 presents the motivation and the focus of the paper. Chapter 2 explores previous works on dynamic traffic light and object detection algorithms. The methodology of this work is presented in Chapter 3. Then, Chapter 4 discusses the experimental settings and the comparison results. The final Chapter 5 concludes the result of this work.

## 2. Related Works

### 2.1. Dynamic Traffic Light

Even though the traffic light has been there for a long time, it needs some changes and upgrades. Previous researchers have researched car detection and how to reduce traffic congestion. Some research studies focus on reducing traffic congestion using car detection and traffic light optimization. Several works of literature have concluded that one of the reasons for traffic congestion is the ineffective traffic light [4, 5, 9, 10].

The traffic light cannot adapt to the current congestion and the vehicles waiting in the intersection [4]. Using the car detection algorithm to detect the car or any vehicle in the intersection [9], the traffic light could adapt to the situation. The traffic light optimization could be supported by a camera in each traffic light to detect vehicles. Then, send signal or data to the system, thus using the calculation to help adapt to the situation and reduce the traffic congestion [4].

The methodology employed for detecting moving vehicles can be likened to that utilized for detecting moving targets [11]. The main goals of this process, which can be categorized into target localization and target classification, are analogous in both vehicle detection and target detection [12]. There exist four primary categories of algorithms for identifying vehicles: those leveraging historical data, those utilizing deep learning techniques, those implementing YOLO, and those tracking a vehicle's trajectory [13].

### 2.2. Object Detection Algorithm

There are several algorithms for object detection, such as Euclidean Distance Tracking [9, 14] and YOLO [15]. Euclidean distance tracking is a tracker that functions as a tracker connected to Euclidean distance tracking to track vehicle movements [9]. After obtaining the center coordinates of each object, the system will create an ID to track the vehicle and save it in the tracking list [14]. On the next frame, Euclidean will re-detect and return the local

coordinates of the detected object and update the Euclidean distance tracking [9]. Euclidean distance integration allows the program to calculate the displacement of any object [14].

The YOLO algorithm architecture is built by 27 CNN layers, with 24 convolutional layers, and then supported by two fully connected layers and a final detection layer [16]. The YOLO algorithm can detect cars in the intersection and help traffic lights adapt to the timing. This will reduce traffic congestion as a dynamic traffic light.

The YOLO series of models for object detection and instance segmentation have undergone rapid development in recent years, with each new version bringing enhancements in precision accuracy [17]. YOLOv8 depicted in Fig. 1, the most recent single-stage model, is based on the groundwork laid by earlier YOLO iterations like YOLOv3 and YOLOv5 [11]. In contrast to two-stage models, YOLOv8 directly forecasts bounding boxes and class probabilities without requiring a separate region proposal network, simplifying the object detection process [17]. A notable advancement in YOLOv8 is the adoption of an anchor-free, center-based approach for object detection, which presents several benefits over conventional anchor-based techniques seen in models like YOLOv5, YOLOv6, and YOLOv7 [12]. YOLOv8 incorporates Pseudo Ensemble or Pseudo Supervision (PS), a technique that entails training multiple models with distinct configurations on the same dataset to produce a more varied range of predictions, thereby enhancing the accuracy and resilience of the final prediction [17]. Furthermore, YOLOv8 makes use of the Darknet 53 architecture, a 53-layer deep convolutional neural network designed for optimized feature extraction and object detection [13].

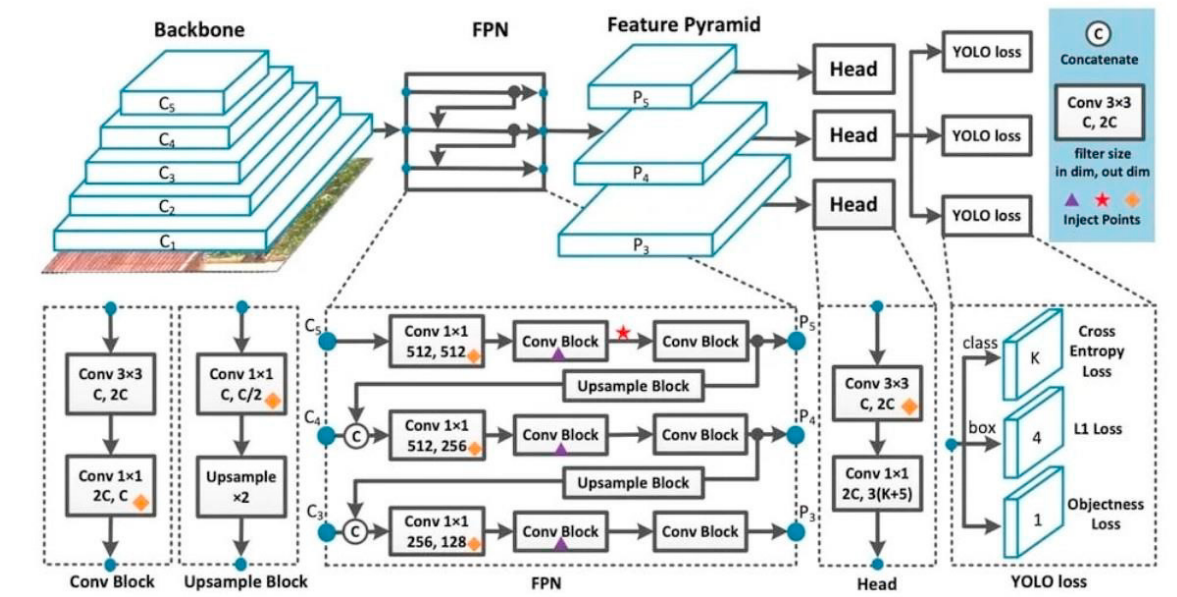


Fig. 1. The innovative design of the YOLOv8 architecture is demonstrated through its application in object detection and segmentation (<https://yolov8.org/what-is-yolov8/>).

### 3. Methodology

The methodology of this work is shown in Fig. 2. There are four main steps: collecting the dataset for the experiment, pre-processing the dataset, testing, and evaluating the results.

#### 3.1. Data Retrieval

The dataset utilized in this work is traffic videos with moving cars from highway CCTV footage. The data is collected from videos on YouTube with the keyword *traffic video for object tracking*. Five videos were carefully selected from the collection. Fig. 3 shows the example of the video.

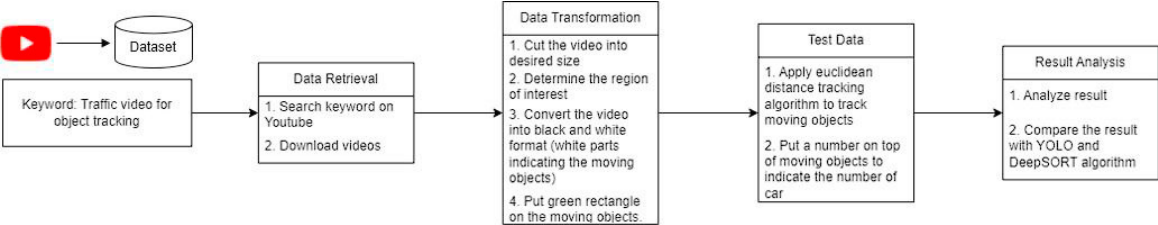


Fig. 2. Research Methodology



Fig. 3. Video capture of vehicle in roadway

3.2. Data Transformation

In the pre-processing step, the video footage was cropped into a 120 x 300-pixel rectangle and placed on the region of interest (ROI). This size was selected to focus on the most relevant area of the footage while minimizing computational overhead. The ROI was selected based on preliminary analysis of typical vehicle movements in the scene, ensuring that the most critical areas for traffic flow analysis were captured. Next, the footage was converted to black and white, with the white part indicating moving objects. This binary representation simplifies the detection of moving objects by creating a clear contrast between static background and dynamic elements. Then, a green rectangle was applied to the white part within the footage. The green rectangle serves as a visual indicator to highlight detected moving objects in the processed footage.

The data will be transformed into the desired format for the Euclidean distance algorithm. The data was transformed into a series of (x, y) coordinates representing the centroids of detected moving objects. The paper only took a specific area of the video for the ROI. Then, apply the mask to the ROI. Fig. 4a shows the ROI of one of the data. The mask will convert the color on ROI to black and white, with white indicating moving objects. Fig. 4b shows ROI with the mask applied. The mask is applied to the ROI using a bitwise AND operation, effectively isolating moving objects within the area of interest.

3.3. Test Data

This work utilized the Euclidean distance and YOLO algorithms to apply object tracking, using default hyperparameters for both models. Euclidean distance tracking algorithm works by finding the distance of the detected object between the current frame and the referenced frame (i.e., the frame before) with the equation (1). If the distance be-

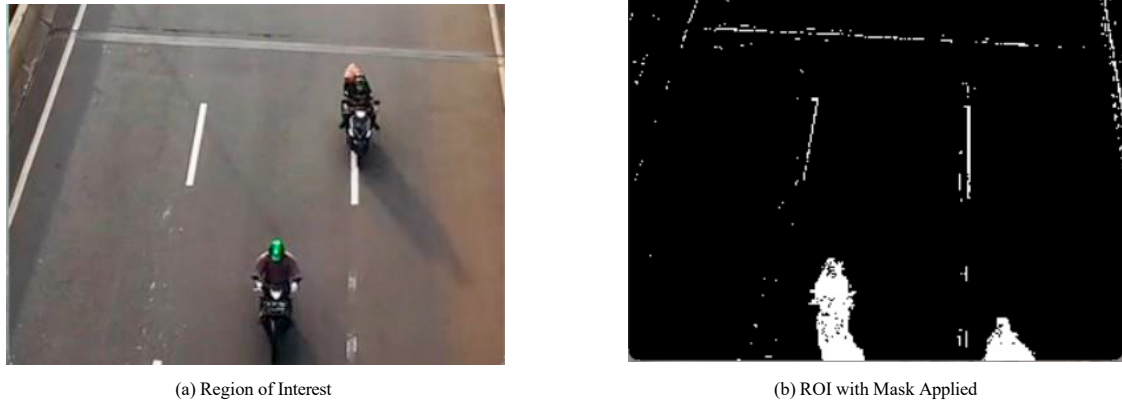


Fig. 4. Image Transformation

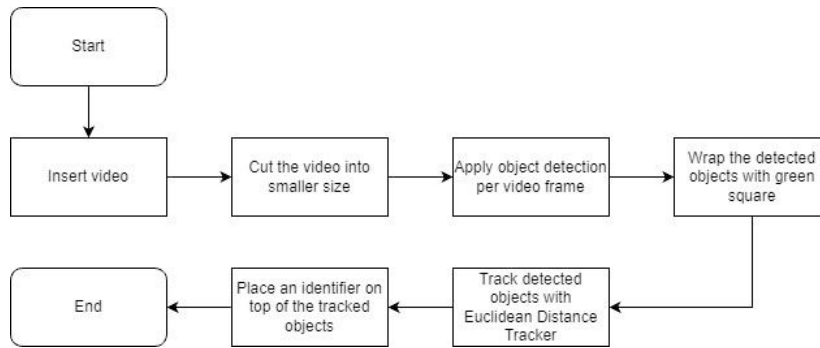


Fig. 5. Euclidean Distance Tracking Process

tween the two objects is less than 50, those objects are considered the same. Fig. 5 above shows the stage of work of the Euclidean distance tracking process. Moreover, the Euclidean pseudocode is shown in Algorithm 1.

$$d(p, q) = \sqrt{(q_1 - p_1)^2 + (q_2 - p_2)^2 + \dots + (q_n - p_n)^2} = \sqrt{\sum_{i=1}^n (q_i - p_i)^2} \quad (1)$$

The YOLO algorithms use a different technique for obtaining vehicle detection counts. They use a neural network to calculate the object location and class in the image. This paper utilizes YOLOv8<sup>1</sup> for the object detection. YOLOv8 can focus on various parts and give better results for this paper's experiments. YOLO uses anchor boxes to pre-define for different sizes and aspect ratios. It is then used to predict the location and size of the object.

The YOLO architecture is shown in Fig. 6. It uses a convolutional neural network (CNN) to detect objects in images. It comprises 53 convolutional layers and has a new technique for improving information flow between layers. The network is divided into the backbone, neck, and head. The backbone is a series of convolutional layers that extract image features. For the neck, it combines features to improve object detection. Lastly, the head is a set of fully connected layers that predict the location and class of the object in the images.

After applying the object detection algorithms, a number was added to the green rectangle to indicate the number of vehicles tracked. Fig. 7 shows the tracked object. In the given example, the algorithm identified two vehicles.

<sup>1</sup> <https://yolov8.com/>

**Algorithm 1** Euclidean Distance Tracking Algorithm

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1: Import necessary libraries
2: Initialize video capture from file
3: Initialize object tracker
4: Create background subtractor for object detection
5: while video is open do
6:   Read a frame from the video
7:   Define the region of interest (ROI)
8:   Apply background subtraction to ROI
9:   Threshold the subtracted image
10:  Find contours in the thresholded image
11:  for each contour do
12:    if contour area is significant then
13:      Calculate bounding rectangle for contour
14:    end if
15:  end for
16:  Update tracker with bounding rectangles
17:  for each tracked object do
18:    Draw bounding rectangle and ID on ROI
19:  end for
20:  Display the frame
21: end while
22: Release video capture
23: Close all windows

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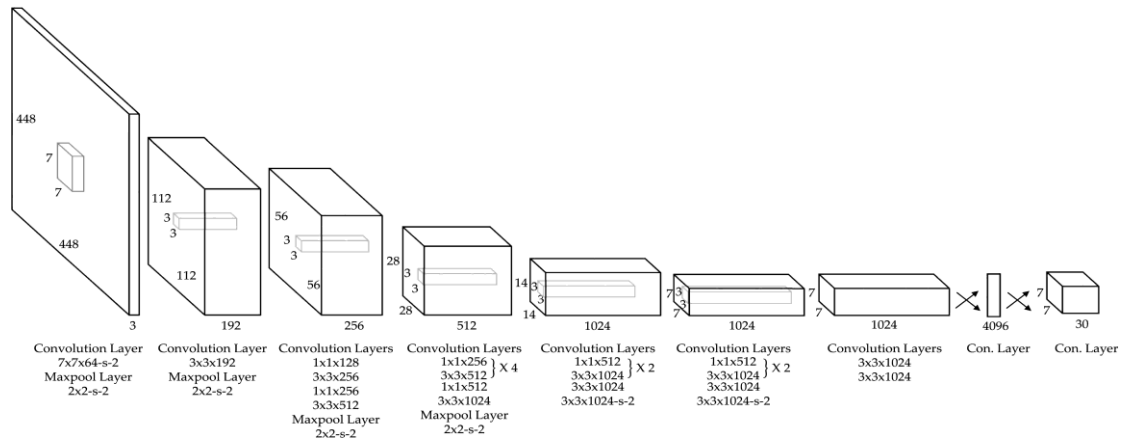


Fig. 6. The architecture for YOLO algorithm [18]

### 3.4. Result Analysis

After obtaining the results from object tracking using the Euclidean distance tracking algorithm, this work compares them with the results of object tracking using the YOLO algorithm on the same video.

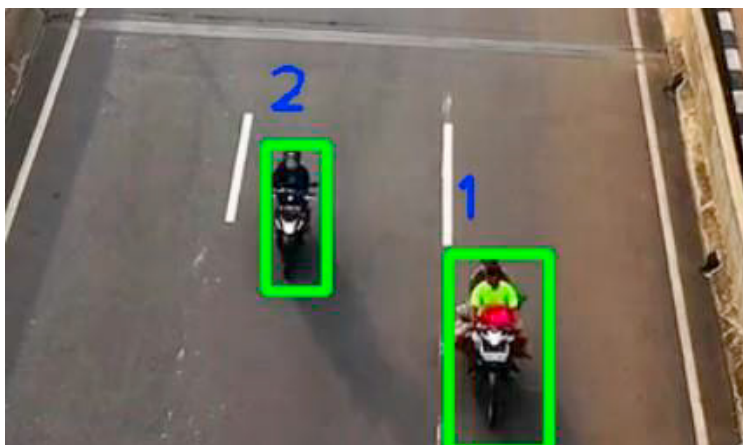


Fig. 7. Tracked Object with Number

## 4. Results and Discussions

### 4.1. Experimental Settings

The hardware specification in this work is a laptop Acer Nitro 5 AN515-57 with Intel (R)Core(TM)i5-11400H CPU, NVIDIA Geforce RTX 3050, and cameras. Moreover, the software and library are Visual Studio Code, Python, Google Colab, and OpenCV.

### 4.2. Results

Using manual counting method for each video, the author was able to get the amount for cars and motorcycles in each video. In the first video, the Euclidean tracked vehicles with a huge error margin of 74, while the actual count was 23. Moreover, Euclidean failed to detect the vehicles for the later experiments, with many inconsistent counts. On the other hand, YOLO succeeded in tracking vehicles with some minor errors. The first video detects 22, while the actual count is 23. Next, the second video had three count differences, and the third video had two count differences. The differences are significant for the fourth video, with an actual count of 34, while the YOLO count is 19. Lastly, the fifth video could not count five cars. In short, while YOLO has some errors, they are better than Euclidean Distance for detecting vehicles. Table 1 shows the result of the experiment.

Table 1: Comparison for Vehicle Count using Euclidean and YOLO

| Video (Vehicle types)    | Vehicle Count | Vehicle Count using Euclidean | Vehicle Count using YOLO |
|--------------------------|---------------|-------------------------------|--------------------------|
| 1 (Cars and Motorcycles) | 23            | 74                            | 22                       |
| 2 (Cars and Trucks)      | 38            | 0                             | 35                       |
| 3 (Cars)                 | 27            | 0                             | 25                       |
| 4 (Cars and Motorcycles) | 34            | 0                             | 19                       |
| 5 (Cars)                 | 13            | 0                             | 8                        |

### 4.3. Discussions

In Table 1, the Euclidean distance tracking algorithm detected 74 vehicles, resulting in a discrepancy of 52 vehicles compared to the actual count. This result has a percentage difference of approximately 236.36% compared to the actual number of vehicles. Conversely, the YOLOv8 algorithm detected 23 vehicles, which is only one vehicle off from the actual count, resulting in a percentage difference of approximately 4.55%. The results from the YOLOv8 algorithm

in the first video are significantly closer to the actual number of vehicles compared to the Euclidean distance tracking algorithm.

The experiment shows that the Euclidean distance tracker is not proficient at tracking moving objects, with many failed tracking for cars, motorcycles, or even trucks. It frequently fails to detect moving vehicles because of failed tracking and the tracker's downsides. Furthermore, Euclidean incorrectly detects the vehicle's parts as a vehicle, such as the windows, rear-view mirrors, and even the vehicle's shadows. This issue arises because the Euclidean distance tracking algorithm determines the center of an object in one frame and compares it with the object's position in the subsequent frame. However, the instability of the camera in this video leads the algorithm to mistakenly identify other objects especially the road's white lines as a moving object and the vehicle's part such as windows and shadow as a different vehicle, thereby disrupting the comparison process. In summary, the Euclidean algorithm cannot be relied on to detect vehicles in traffic. On the other hand, YOLO can fairly well track vehicles with minor errors.

This study indicates several limitations in accurately tracking and counting vehicles using the current system and video footage. Some challenges that are encountered on the experiments result. For instance, the tracking performance suffered in scenarios involving camera movement or multiple vehicle types. The process of calculating comparison between manual method and algorithm, such as YOLO and Euclidean Distance poses some challenges, especially when dealing with large number of vehicles. During the experiment, the Euclidean Distance Algorithm frequently shows lack of consistency and high failure rate, causing numerous repetitions and resulting in experimental failures.

These show that Euclidean Distance Algorithm is not fit to be used as an algorithm for vehicle detection. While Euclidean Distance algorithm has faster time in detecting vehicles, the result is inconsistent and not accurate. In contrast, YOLO is superior in accuracy and consistency with lower margin error.

To achieve more consistent results, utilizing the YOLO algorithm is more recommended. For successful implementation of the algorithm, some conditions need to be achieved, the camera movement have to be minimized and need to be more stable. Camera stability improves the accuracy and consistency of vehicle tracking systems. Further advancements in object recognition algorithms and stability against camera movement are necessary to improve the accuracy and consistency of vehicle tracking systems.

## 5. Conclusion

The algorithms used in this work are Euclidean Distance and YOLO. Both of these algorithms could detect vehicles, but the results differ. The experiment shows that Euclidean distance has a hard time detecting moving vehicles in the given video, with a considerable margin of error in the calculation, and could also detect traffic lines or moving objects other than vehicles. On the other hand, YOLO could detect every vehicle more accurately than Euclidean and give better results with some errors.

For future work, Euclidean distance still needs more tweaks and upgrades for vehicle detection. Background subtraction can be utilized to differentiate moving objects from the video footage. Applying frame skipping might also help improve the Euclidean algorithm's performance.

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